Math 340
Fall 2021
Final Exam
12/22/2021
Time Limit: 120 Minutes

Name (Print): ____________________

TA's name and Section#: ____________________

Student ID Number: ____________________

This exam contains 10 pages (including this cover page) and 12 problems. The last page is the scratch paper. If any pages are missing, please inform the instructor immediately.

Directions:

- Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.
- You may *not* use your calculators, books and notes.
- You must show all your work. A stranger should be able to pick up your paper and follow your reasoning. You must use proper notation and define the events clearly. Partial credits may be given.
- Unless otherwise noted, your answers must be in fraction form or terminating decimals.
- Circle your answer to the True/False problems and justify your answer.
- For the free response problems you need to show all necessary work in the space provided for each problem. Answers must be justified with sufficient work, and partial credit will be given for appropriate work shown.

I do not lie, cheat or steal, or tolerate those who do.

By signing below you indicate that all your work is your own and that you have neither given nor received help from external sources.

Signature: ____________________ Total: ____________________

T/F		P. 9	
P.6		P. 10	
P.7		P. 11	
P.8		P. 12	

True/False Questions: If true, give your justification. If false, give a counterexample or some reasoning.

1. (2 points) Let V be a vector space of dimension 4. If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is a basis of V, then so is $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \mathbf{v}_3 + \mathbf{v}_4, \mathbf{v}_4 + \mathbf{v}_1\}$.
   A. True.
   B. False.

2. (2 points) If $\mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{w}$, then $\mathbf{v}$ is orthogonal to every vector of the space spanned by $\{\mathbf{u}, \mathbf{w}\}$.
   A. True.
   B. False.

3. (2 points) Let A be an $n \times n$ matrix. If A is singular, then 0 is an eigenvalue of A.
   A. True.
   B. False.

4. (2 points) ($\star$) Let A be any square matrix. Then the matrix $A + A^T$ is always diagonalizable.
   A. True.
   B. False.

5. (2 points) ($\star$) All eigenvalues of a symmetric real matrix are real and positive.
   A. True.
   B. False.

6. (10 points) Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix} \text{ and } B^{-1} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}.$$

(a) Compute $\det(A)$.

(b) Compute $\det(B)$.

(c) Compute $(A^{-1}B^T)^{-1}$.

7. (10 points) Let $L : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by

$$L\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_3 \\ 2x_1 + x_2 + x_3 \\ 3x_1 + x_2 \end{bmatrix}.$$

(a) Find Ker$L$.

(b) Is L onto? Give your reason.

(c) Is L invertible? If so, find $L^{-1}\left(\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}\right)$.

8. (10 points) Let A be the following matrix

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

(a) Compute the characteristic polynomial of A.

(b) Determine the eigenvalues of A and their algebraic multiplicities.

(c) Is A diagonalizable? If so, find a diagonalization (That is, find matrices D and P such that $A = PDP^{-1}$). If not, explain why.

9. (10 points) ($\star$) Find an orthonormal basis for the column space (i.e., the subspace spanned the column vectors) of the matrix:

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}.$$

10. (10 points) Let P_2 be the space of polynomials with $\mathbb{R}$-coefficients of degree ≤ 2. We define an inner product on P_2 by
$$(f, g) = \int_0^1 f(t)g(t)dt.$$

(a) Let $g_1 = t - 1$ and $g_2 = t^2$. Let θ be the angle between g_1 and g_2. Compute the norms (or lengths) of g_1, g_2 and $\cos(\theta)$.

(b) ($\star$) Compute the closest distance between g_2 and vectors in the span of g_1.

11. (10 points) Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$L\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}\right) = \begin{bmatrix} 2u_1 + u_2 \\ u_1 + 3u_2 \end{bmatrix}.$$

Let $S = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}\right)$ be the ordered basis for $\mathbb{R}^2$. For this question, evaluate all matrix inverses and products.

(a) Find the matrix representation of L with respect to the natural basis $E = \{\vec{e}_1, \vec{e}_2\}$ for $\mathbb{R}^2$.

(b) What is the transition matrix $P_{S \leftarrow E}$ from the E-basis to the S-basis?

(c) Find the matrix representation of L with respect to the ordered basis S.

12. (10 points) Let $L : P_2 \to P_2$ be defined by

$$L(at^2 + bt + c) = (a + (1 - \lambda)b)t^2 + (b + c)t + (a + (1 - \lambda^2)c)$$

(a) Represent L by a matrix with respect to the standard basis $(t^2, t, 1)$ (Hint: Only λ should appear in the matrix as a variable.)

(b) For which values of λ the transformation is invertible?

(c) When $\lambda = 1$, find all polynomials $f \in P_2$ such that $L(f) = 2t^2 + 3t + 2$.

Scratch Paper